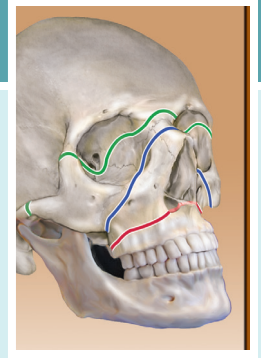


Team Medical Coverage

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The roots of peak athletic performance are embedded in the optimal health of athletes. It is the responsibility of every member of the sports medicine team to provide consistent, thorough, evidence-based, and comprehensive care for each athlete. Team medical coverage has evolved rapidly and has become a pivotal component of athletics at all levels of competition. The ultimate responsibility for medical decisions regarding both prevention and treatment rests with a team of professionals, which includes the medical director, other team physicians, athletic trainers, and allied health care providers.¹ The minimum qualifications for the medical director and other team physicians include¹:

1. Having a Doctor of Medicine (MD) or a Doctor of Osteopathy (DO) degree in good standing, with an unrestricted license to practice medicine.
2. Possessing a fundamental knowledge of emergency sport-related medical care.
3. Being trained in cardiopulmonary resuscitation (CPR) and the use of an automated external defibrillator (AED).
4. Having a working knowledge of musculoskeletal injuries, medical conditions, and psychological issues affecting athletes.

Duties of the medical team are numerous but primarily include: (1) effectively coordinating preparticipation evaluations, (2) managing on-field emergencies, and (3) navigating the ethical and medicolegal issues unique to sports medicine. These topics are discussed in detail in this chapter.

PREPARTICIPATION PHYSICAL EVALUATION

A mainstay of modern medicine is the idea of preventive health care. In the realm of organized athletics, the preparticipation physical evaluation (PPE) has historically served as the cornerstone of the prevention of unnecessary morbidity and mortality in sport. In May 2010, recommendations for the PPE were updated and published (PPE-4).² Based on expert consensus and a comprehensive review of the medical literature, the PPE-4 was developed with input from six medical societies and is endorsed as the “gold standard” for the PPE.² However, historically, PPE standards in the United States have been determined by individual state and local legislative authorities, with varying medical and legal guidelines and community resources. As a result, the implementation of the PPE nationally has been heterogeneous, with variable compliance by educational institutions and athletic associations. The lack of uniform application of the PPE has made it difficult to interpret outcome data for the few studies

that exist.³ A review found no medium- or better-quality evidence that the PPE reduces morbidity or mortality.⁴ Despite variation in the format and use of PPEs, it is held that, at the very least, the PPE allows for establishment of a “medical home,” an immunization record, identification and management of acquired and congenital medical conditions, and proactive counseling related to medical conditions in sports and lifestyle risk factors.⁵

Goals and Objectives

Many sports governing bodies mandate a PPE not only to ensure the health and well-being of the participant but also to minimize legal liability for conditions that may occur or worsen with participation in sports.⁵ The principal goal of the author societies who formulated the PPE-4 is the promotion of the health and safety of athletes.

Primary objectives of the PPE include²:

1. Screening for potential life-threatening or disabling conditions.
2. Screening for conditions that may predispose to injury or illness.

Secondary objectives that address general health care and prevention² include:

1. Determining general health.
2. Serving as an entry point to the health care system.
3. Providing an opportunity to initiate a discussion on health-related topics.

The PPE may be the only annual medical appointment that athletes have. It is thus important to take advantage of this valuable time to initiate a discussion on drug use, seat belt and helmet use, and safe sexual practices. Failure to limit risk-taking behaviors is more likely to cause harm in adolescents than is participation in sports.⁶

Timing, Setting, and Organization

Ideally, the PPE should take place at least 6 weeks before the start of organized practices. A small percentage of athletes may be conditionally withheld from participation pending further workup or consultation with a specialist, or they may have ongoing medical issues that need to be resolved. In general, more frequent evaluation is recommended for younger athletes because cardiac abnormalities may manifest during dynamic growth periods, despite normal results of earlier evaluations. At the secondary school level, comprehensive examinations should occur every 2 to 3 years, interspersed with problem-focused evaluations annually. For the collegiate age group, it is recommended that

Abstract

Peak athletic performance is dependent on the optimal health of athletes. The preparticipation physical evaluation is an integral component of achieving this underlying goal of maintaining optimum health for athletes and to prevent unnecessary morbidity and mortality in sport. This chapter addresses specific details of each element of the preparticipation physical evaluation for healthy athletes and for athletes with special needs. In addition, emergency action plans are discussed for on-field emergencies such as head injury, cervical spine injury, cardiac arrest, tension pneumothorax, anaphylaxis, and limb threatening injuries. Environmental factors such as heat illness and injuries caused by lightning are also covered in detail. Finally, bioethics will be discussed, including sample cases depicting issues of confidentiality and informed consent.

Keywords

preparticipation physical exam
on-field emergencies
bioethics

a comprehensive evaluation be performed prior to the first season and that shorter, more focused evaluations be performed annually thereafter. A new recommendation by the PPE-4 is for children (>6 years) and adolescents to undergo a PPE as part of their annual health maintenance examination. The authors of the PPE-4 reason that all children should be encouraged to be active, given recent trends in obesity and inactivity. Children who plan to be active in nonstructured sport settings are equally susceptible to injury and illness and thus should be offered a PPE.²

The two typical approaches to the PPE are (1) a primary care physician's office visit and (2) coordinated team examinations. A comparison of the two settings is outlined in Table 11.1. The PPE-4 recommends examinations at the primary care physician's office. During team examinations, the ultimate responsibility to clear athletes for sport participation is incumbent upon a physician with an unrestricted MD or DO license.² A standardized history questionnaire should be completed by the athlete prior to arrival, and parental guidance is advocated for minors. The PPE-4 consistently reinforces the adage that the history is the most important diagnostic tool. It has been shown that 75% or more of medical and orthopedic conditions are detected by history alone.⁷⁻⁹ The PPE-4 History and Physical Exam forms are free and are available online.¹⁰

History and Physical Examination

Head, Ears, Eyes, Nose, and Throat

A detailed history of eye disorders, injuries, and the need for corrective or protective eyewear should be obtained. Visual acuity should be assessed via a standard Snellen eye chart. Poor visual acuity is one of the most frequently reported findings on PPEs, and appropriate consultation with a specialist should be facilitated.¹¹ Athletes whose visual acuity in the poorer eye is worse than 20/40 with optical correction should be classified as being "functionally one-eyed," and akin to persons missing an eye, they should wear protective eyewear when participating in sports with a high risk for eye injury. Furthermore, they should be

restricted from participating in sports that feature intentional harm, such as full-contact martial arts, boxing, and wrestling.² A pupillary examination to document baseline anisocoria should be performed for reference in the event of a head injury. Odon-togenic pathology such as oral ulcers, gingival atrophy, and decreased enamel may be a sign of an eating disorder.²

Cardiovascular System

Sudden cardiac death (SCD) is the most feared complication of sports participation. The annual incidence of SCD in the United States is approximately 1 in 80,000 high school athletes and 1 in 50,000 college athletes.¹² Box 11.1 outlines the most common causes of sudden death in young athletes in the United States. To reduce the risk of SCD during sports participation, the PPE-4 endorses the recommendations made by the 2007 American Heart Association (AHA) consensus statement on preparticipation cardiovascular screening.¹³ According to multiple studies, the sensitivity of history taking and a physical examination for detecting cardiac disorders associated with SCD is 20%.^{14,15} The author societies of PPE-4 admit that no outcome-based studies have demonstrated that the PPE is effective for preventing or detecting athletes at risk for SCD.² The PPE-4 questions used to evaluate cardiac history (personal and family) are listed in Fig. 11.1. These questions have been expanded in the PPE-4 to help identify ion channelopathies such as long QT and Wolff-Parkinson-White syndromes.

Current AHA guidelines recommend that the standard 11-point screening guideline and those of the PPE-4 are used by examiners as part of a comprehensive history and physical examination to detect cardiovascular abnormalities for high school and college athletes in the United States.¹⁶ The physical examination recommended by the AHA includes¹⁷ cardiac auscultation for heart murmurs,¹⁸ palpation of radial and femoral pulses to assess for aortic coarctation¹⁹ examination for physical stigmata of Marfan syndrome (MFS),¹⁹ and brachial artery blood pressure taken in the seated position.¹³ Cardiac auscultation should be performed with athletes seated and also with Valsalva maneuver and squatting. The murmur related to hypertrophic cardiomyopathy (HCM) will be accentuated by the Valsalva maneuver and decreased with squatting, whereas the murmur related to aortic stenosis would follow the opposite pattern. Soft early systolic murmurs are common in athletes and represent hyperdynamic flow rather than anatomic pathology.² Blood

TABLE 11.1 Comparison of Preparticipation Physical Evaluation Clinical Settings

Office Based	Coordinated Team
Pros	Pros
Better continuity of care	Time efficient
Established relationship	Reduced cost
Privacy	Available to athletes without a primary care provider
Access to medical records	Immediate access to specialists and athletic trainers
Allows time for counseling	
Cons	Cons
Time intensive	Lack of privacy
Potential need for specialist referral	Decreased time for counseling
Increased cost	Success is dependent on the ability of the athlete or parent to accurately complete a questionnaire

BOX 11.1 Common Causes of Sudden Death in Young Athletes

- Hypertrophic cardiomyopathy
- Commotio cordis
- Coronary anomalies
- Left ventricular hypertrophy of indeterminate cause
- Myocarditis
- Ruptured aortic aneurysm
- Arrhythmogenic right ventricular cardiomyopathy
- Heat stroke

Modified from Maron BJ. Sudden death in young athletes. *N Engl J Med.* 2003;349(11):1064–1075.

HEART HEALTH QUESTIONS ABOUT YOU	Yes	No
5. Have you ever passed out or nearly passed out DURING or AFTER exercise?		
6. Have you ever had discomfort, pain, tightness, or pressure in your chest during exercise?		
7. Does your heart ever race or skip beats (irregular beats) during exercise?		
8. Has a doctor ever told you that you have any heart problems? If so, check all that apply: ☐ High blood pressure ☐ A heart murmur ☐ High cholesterol ☐ A heart infection ☐ Kawasaki disease Other: _____		
9. Has a doctor ever ordered a test for your heart? (For example, ECG/EKG, echocardiogram)		
10. Do you get lightheaded or feel more short of breath than expected during exercise?		
11. Have you ever had an unexplained seizure?		
12. Do you get more tired or short of breath more quickly than your friends during exercise?		
HEART HEALTH QUESTIONS ABOUT YOUR FAMILY	Yes	No
13. Has any family member or relative died of heart problems or had an unexpected or unexplained sudden death before age 50 (including drowning, unexplained car accident, or sudden infant death syndrome)?		
14. Does anyone in your family have hypertrophic cardiomyopathy, Marfan syndrome, arrhythmogenic right ventricular cardiomyopathy, long QT syndrome, short QT syndrome, Brugada syndrome, or catecholaminergic polymorphic ventricular tachycardia?		
15. Does anyone in your family have a heart problem, pacemaker, or implanted defibrillator?		
16. Has anyone in your family had unexplained fainting, unexplained seizures, or near drowning?		

Fig. 11.1 Cardiac history screening. (From Bernhardt DT, Roberts WO, eds. *Preparticipation Physical Evaluation*, 4th ed. Elk Grove Village, IL: American Academy of Pediatrics; 2010.)

pressure measurements should be made with an appropriately sized cuff. All athletes with hypertension require further workup and monitoring. Athletes with stage I hypertension (i.e., the 95th to 99th percentile plus 5 mm Hg in children and 140/90 to 160/100 mm Hg in adults) who do not have heart disease, end-organ damage, or left ventricular hypertrophy should not be restricted, except from sports that may make heavy use of the Valsalva maneuver, such as weightlifting. Athletes with stage II hypertension (i.e., >99th percentile plus 5 mm Hg in children and >160/100 mm Hg in adults) should initially be restricted from all participation pending a thorough workup and control of blood pressure.²⁰

Special attention should be directed at excluding MFS (and related connective tissue disorders, such as Ehlers-Danlos and Loeys-Dietz syndromes), which all manifest genetic deficiencies of connective tissue proteins and thereby increase risk of dissection of the aorta and other smaller arteries. Clinical manifestations of MFS may also include the ocular, musculoskeletal, respiratory, neurologic, and integumentary systems (Box 11.2). For the diagnosis of MFS, revised Ghent criteria were published

BOX 11.2 Phenotypic Features of Marfan Syndrome

Cardiac

- Aortic root dilation
- Mitral valve prolapse

Ocular

- Ectopia lentis
- Myopia

Musculoskeletal

- Tall stature (males taller than 183 cm [6 feet] and females taller than 178 cm [5 feet, 10 inches]); *Note: Tall stature is not formally considered a criterion of the revised Ghent nosology*
- Increased upper segment to lower segment ratio; increased arm span relative to height
- Arachnodactyly (wrist and thumb sign)
- Pectus carinatum/excavatum
- Hindfoot deformity or plain pes planus
- Protrusio acetabuli
- Scoliosis, kyphosis
- Reduced elbow extension

Facial Features

- Dolichocephaly
- Enophthalmos
- Downslanting palpebral fissures
- Malar hypoplasia, retrognathia

Other Systemic Features

- Pneumothorax
- Dural ectasia
- Skin striae

Modified from Loeys BL, Dietz HC, Braverman AC, et al. The revised Ghent nosology for the Marfan syndrome. *J Med Genet*. 2010;47(7):476–485; Giese EA, O'Connor FG, Brennan FH, Depenbrock PJ, Oriscello RG. The athletic preparticipation evaluation: cardiovascular assessment. *Am Fam Physician*. 2007;75(7):1008–1014.

in 2010 and consequently are not referenced in the PPE-4.²¹ The revised Ghent criteria have placed more weight on the three cardinal features of MFS¹⁷: aortic root enlargement,¹⁸ ectopia lentis,¹⁹ and FBN1 mutation to differentiate the manifestations of the cardiovascular, ocular, and skeletal organ systems. In the absence of family history, the presence of both aortic root changes and ectopia lentis¹⁹ are sufficient to make the diagnosis. However, these two features are not easily assessed in the PPE setting, and thus clinicians should be vigilant to identify any possible physical stigmata of MFS, particularly in male athletes taller than 183 cm (6 feet) and female athletes taller than 178 cm (5 feet, 10 inches). Athletes with a family history of MFS or two or more of the aforementioned physical examination findings should be referred for cardiology consultation to comprehensively assess aortic morphology.²² Given the greater emphasis on ectopia lentis in the revised criteria, a referral to ophthalmology for slit-lamp testing is also prudent.²¹

Significant debate exists regarding noninvasive cardiac screening during the PPE, and although comprehensive coverage of this

subject is beyond the scope of this chapter, related controversies are discussed in detail elsewhere in this book. The current AHA guidelines limit the use of electrocardiography (ECG) screening to small cohorts of young people aged 12 to 25 years with suspected cardiovascular pathology based on history and physical examination, provided that close physician supervision and quality control are achieved.¹⁶ The guidelines suggest that limitations of using a 12-lead ECG, including poor sensitivity and specificity, should be anticipated. Otherwise, current guidelines do not agree on systematic inclusion of ECG in preparticipation screening of athletes and nonathletes to identify congenital or other cardiovascular abnormalities.^{16,23} These guidelines are in contradistinction with recommendations of the International Olympic Committee²⁴ and the European Society of Cardiology (ESC),²⁵ which recommend the use of a screening ECG during PPEs. Routine ECG screening is supported by Italian data showing a 90% decrease in SCD rate with the inclusion of ECG in PPE cardiovascular screening.¹⁴ The dispute in recommendations among societies is largely based on studies performed in Italy^{26–28} which demonstrated that an ECG had a 77% greater power to detect HCM than did a history and physical examination alone.²⁸

There are important sex and race differences in SCD rates that may influence the decision to use the ECG as part of the PPE. It has been shown that there is an increased likelihood of obtaining abnormal ECGs with PPEs of older adolescent males and a decreased likelihood with females.²⁹ Male athletes have been shown to be at 5 to 6 times greater risk than female athletes for SCD.³⁰ In addition, the rate of SCD among black athletes has been shown to be 3 to 5 times higher in comparison with all other ethnic groups.³⁰

A systematic approach to interpreting screening ECGs as part of the PPE are the Seattle Criteria.³¹ These criteria were initially created to balance the sensitivity and specificity of ECG use, while maintaining a practical and concise checklist for physicians to use for ECG interpretation with athletes. The goal of the Seattle Criteria is to aid physicians in distinguishing ECG abnormalities that may be considered normal in athletes from abnormal ECG findings that are concerning for pathology. However, use of the Seattle Criteria has not been shown to decrease morbidity or reduce incidence of SCD among athletes.^{31–33} The Refined Criteria published in 2014 for ECG interpretation have been shown to decrease the occurrence of false-positive and false-negative results.³³ Data suggest that the Refined Criteria outperform the ESC recommendations and Seattle Criteria by reducing false-positive rates of ECG interpretation in black athletes from 40.4% (ESC) and 18.4% (Seattle Criteria) to 11.5%, and in white athletes from 16.2% (ESC) and 7.1% (Seattle Criteria) to 5.3%.^{33,34} These studies suggest that established criteria and standardized guidelines for ECG interpretation are useful to avoid false-positive interpretations and potentially costly and unnecessary additional workup. Recently, the American Medical Society for Sports Medicine (AMSSM) formed a task force to address the need for preparticipation cardiovascular screening in young competitive athletes. Although the AMSSM supported the position that the ECG increases early detection of cardiac disorders associated with SCD, the absence of clear outcome-based research precluded the endorsement of using the ECG as a universal cardiovascular

screening strategy.¹⁵ If physicians choose to incorporate routine ECG screening into their PPEs, proficiency in accurate ECG interpretation and access to cardiology resources are of paramount importance to maximize its utility without unnecessary cost escalation. However, the use of ECG as a global cardiovascular preparticipation screening modality in athletes is limited by the low prevalence of SCD in athletes.

A newer area of debate is use of an echocardiogram as part of the routine PPE cardiac screening protocol. It has been demonstrated that ECGs have a higher rate of false-positives as a screening modality in comparison with echocardiogram for detection of causative risk factors for HCM.¹⁷ This is in large part due to the similarities between ECG changes that are physiologic resulting from physical conditioning and the structural and electrical changes found in HCM. The increasing availability of portable ultrasound has led to the development of the Early Screening for Cardiac Abnormalities with Preparticipation Echocardiography (ESCAPE) protocol, which produced accuracy and reliability in diagnosing results indicative of HCM and aortic root dilation.³⁵ The protocol demonstrated that limited echocardiography measurements by noncardiologists were not statistically different from values obtained from cardiologists trained in echocardiography interpretation.^{34,35} However, as with ECG interpretation, the ability of the team physician, not frequently a cardiologist, to accurately and efficiently interpret echocardiograms is paramount to this modality being as useful as is reported in the literature.

Reducing the incidence of SCD is a noble goal, but clinicians should be aware of the high false-positive rates associated with PPE ECGs, along with high costs, increased use of medical resources, potentially erroneous disqualification of athletes from participation, and the potential for lost “life-years” following needless sedentary lifestyle recommendations.³⁶ Therefore although controversial, at present there are insufficient data available to promote universal use of screening ECGs or echocardiograms in asymptomatic individuals for detecting cardiovascular disease. A rational decision to include ECG or echocardiography as part of the PPE could conceivably be made in the appropriate setting and with sufficient provider training and resources.

Pulmonary System

It is imperative that athletes be questioned regarding a personal or family history of asthma or exercise-induced bronchoconstriction (affecting 50% to 90% of athletes), asthma-like symptoms and their severity, use of bronchodilators, degree of asthma control, and the use of tobacco or exposure to secondhand smoke.³⁷ A thorough auscultation of the lung fields should be performed during the PPE. Proper diagnosis may require provocative pulmonary function testing, and in patients with exercise-induced bronchospasm, this is imperative to evaluate for undiagnosed asthma.³² Because an athlete's symptoms may be related to the environment, these tests may need to be performed in environmental conditions that stimulate symptoms.³⁷ Athletes who are recovering from an asthma exacerbation and who are actively wheezing must be restricted from sports activity until stabilized.³² Vocal cord dysfunction should be considered as a potential diagnosis in athletes with asthma-like symptoms who fail to respond

to usual bronchodilator therapy. Athletes with stable asthma or exercise-induced bronchoconstriction can usually be cleared to participate in sports, unless they are recovering from a recent asthma exacerbation. It may be indicated for athletes with severe underlying asthma or pulmonary disease to have a rescue inhaler available as a condition for athletic participation.³²

Gastrointestinal/Genitourinary System

The abdomen should be thoroughly examined for masses, tenderness, rigidity, or enlargement of the liver or spleen. The AMSSM, among other sports medicine professional organizations, recommends that a detailed genitourinary examination should be performed to assess for masses, testicular descent, tenderness, and hernias in male athletes.² Liver or spleen enlargement is a contraindication to participation in sports. The athlete with a solitary kidney or testicle may be cleared to participate in contact sports; however, each athlete should be counseled on an individual basis about the harmful consequences of injury, and appropriate protective gear should be mandated during practice and competition. Acute diarrheal illness is a contraindication to participation in sports unless symptoms are mild and the athlete remains well hydrated.²

Musculoskeletal System

The musculoskeletal history has a very high sensitivity (92%) for detecting abnormalities, and thus the physician should inquire about current injuries and a history of injuries requiring evaluation, casting, bracing, surgery, or missed participation.^{5,38} Given the high sensitivity of the history, a screening musculoskeletal examination is sufficient but should be supplemented with a more in-depth joint-specific examination when pathology is suspected.² **Box 11.3** reviews the 11-step general musculoskeletal screening examination. In general, literature suggests that generalized joint laxity, imbalance in strength ratios, excessive foot pronation or supination, level of physical maturity, multidirectional

balance, and body mass all may be predictive for risk of injury in the adolescent athlete population.^{39,40} The PPE-4 recommends functional testing with the duck walk and single-leg hop as part of the musculoskeletal examination.² These physical examination tests are time-efficient, require no additional equipment, and may help clinicians to effectively assess multiple physical attributes during the PPE.

Clearance for sport participation is determined based on the degree and type of injury or condition, the ability of the injured athlete to compete safely, and the necessities of a given sport. In general, if the athlete has no tenderness to palpation, has normal range of motion (ROM) and normal strength in the affected area, and performs adequately on functional tests, clearance for sport participation can typically be given. Use of protective padding, taping, or bracing may also enable the athlete to participate in sports safely.²

Neurologic System

Physicians should inquire about prior concussions or head injuries, seizure disorders, frequent or exertional headaches, problems with recurrent burners or stingers (transient brachial plexopathy), or a prior transient quadriparesis or cervical cord neuropraxia.² A positive history demands a thorough neurologic examination, including assessment of cognition, cranial nerves, motor-sensory function, muscle tone, reflexes, coordination testing, and gait evaluation.⁵ Athletes who have a history of multiple concussions or who have prolonged postconcussive symptoms should be counseled about risks and encouraged to discuss with their families the potential for harm, although not presently well understood, that may result from repeated concussions.⁵ The Sport Concussion Assessment Tool (SCAT),⁴¹ exercise testing such as the Buffalo Concussion Treadmill Test,⁴² vestibuloocular system testing such as the Vestibular/Ocular Motor Screening Assessment,⁴³ vestibulospinal testing such as the Balance Error Scoring System,³² and neuropsychological testing⁴⁴ may increase the sensitivity of detecting residual concussion symptoms. Athletes with persistent symptoms should be disqualified and cleared for participation only after they have successfully completed a graded return-to-sport protocol without recurrence of concussion-related signs or symptoms.⁴¹ Likewise, after sustaining a single stinger, athletes must be asymptomatic with a normal neurologic examination prior to medical clearance, although further diagnostic testing may be necessary for persistent symptoms or recurrent injury.⁴⁵ Athletes who have had transient quadriparesis should have an MRI to evaluate for spinal stenosis.⁴⁶ For atypical, exercise-related headaches, advanced intracranial imaging may be necessary to evaluate for occult causes of secondary headaches.⁴⁷ Although a history of seizures does not preclude athletic participation, sports-specific modifications may be needed, particularly for persons involved in water sports.²

BOX 11.3 General Musculoskeletal Screening Examination

The general musculoskeletal screening examination consists of the following: (1) inspection, athlete standing, facing toward examiner (symmetry of trunk, upper extremities); (2) forward flexion, extension, rotation, lateral flexion of neck (range of motion, cervical spine); (3) resisted shoulder shrug (strength, trapezius); (4) resisted shoulder abduction (strength, deltoid); (5) internal and external rotation of shoulder (range of motion, glenohumeral joint); (6) extension and flexion of elbow (range of motion, elbow); (7) pronation and supination of forearm or wrist (range of motion, elbow and wrist); (8) clench fist, then spread fingers (range of motion, hand and fingers); (9) inspection, athlete facing away from examiner (symmetry of trunk, upper extremities); (10) back extension, knees straight (spondylolysis/spondylolisthesis); (11) back flexion with knees straight, facing toward and away from examiner (range of motion, thoracic and lumbosacral spine; spine curvature; hamstring flexibility); (12) inspection of lower extremities, contraction of quadriceps muscles (alignment, symmetry); (13) "duck walk" 4 steps (motion of hip, knee, and ankle; strength; balance); (14) standing on toes, then on heels (symmetry, calf; strength; balance).

From Bernhardt DT, Roberts WO, eds. *Preparticipation Physical Evaluation*. ed 4. Elk Grove Village, IL: American Academy of Pediatrics; 2010.

Hematologic and Infectious Disease

The risks of transmission of human immunodeficiency virus, hepatitis B, and hepatitis C during routine sports participation are considered minimal, and the presence of these infections is not considered a contraindication to participation.^{48,49} However, guidelines from the National Hemophilia Foundation advise

that athletes with bleeding disorders such as hemophilia be restricted from contact or collision sports. In addition, a diagnosis of infectious mononucleosis should preclude all sports participation for the first 3 weeks because of a significantly elevated risk of splenic rupture.⁵⁰ Light, noncontact activity may be recommended 3 weeks after diagnosis if the athlete is asymptomatic and without complications. Participation in full-contact sports should be deferred for at least 28 days from the start of symptoms or from clinical diagnosis if the symptom onset date is imprecise.² Febrile illness is also a contraindication to participation in sports because of increased susceptibility to heat illness and because it may occasionally accompany conditions such as myocarditis that may make sports participation unsafe.²

Sickle cell trait (SCT) is associated with 2% of deaths in National Collegiate Athletic Association (NCAA) football players.⁵¹ Athletes should be questioned about a personal or family history of SCT or sickle cell disease (SCD). Phenotypic expression is varied, and thus medical clearance or exercise modifications should be made on an individual basis. To prevent exertional sickling collapse, athletes with SCT should avoid strenuous activity in extreme heat and high-altitude environments, especially when they are poorly acclimated.⁵² Deaths from exertional sickling have reportedly been more common among football players as well.⁵³ In recognition of these risks, the NCAA in 2010 mandated that all Division I and II athletes be screened for SCT and that the status be established at the time of the PPE³²; however, athletes may decline screening.⁵¹ This requirement by the NCAA was extended to Division III competitors as of 2014.⁵⁴ If SCT is confirmed with screening, affected athletes are offered counseling on its implications to their health, athletics, and family planning.⁵⁵ Debate still exists on universally screening athletes for SCT. Some proponents have estimated that universal screening of NCAA Division I student athletes would save seven lives over 10 years.⁵⁴ Based on the available evidence, we recommend that all active individuals be screened for SCT with a 1-time hemoglobin electrophoresis test.

Endocrine System

Persons with types I and II diabetes mellitus can participate in sports without restriction but are encouraged to monitor blood glucose more frequently, maintain a balanced diet, adjust medications appropriately, and hydrate suitably.² Education regarding activities with relatively higher risk of cutaneous foot injury such as hiking, rock climbing, or surfing is also recommended. Obese patients should not be discouraged from participating in sports but should receive counseling on lifestyle changes such as dietary and activity modifications, as well as prevention of heat-related illness.² Athletes in weight-sensitive sports such as wrestling and boxing, and aesthetic sports such as diving and figure skating are at risk for eating disorders, particularly in females.³² The female athlete triad consists of disordered eating, amenorrhea, and decreased bone mineral density. A history of stress fractures with prolonged healing in a female athlete should increase clinical suspicion for this treatable yet potentially deadly medical condition. These patients should undergo a multidisciplinary treatment program with further risk stratification before they return to sports. It is recommended that sports medicine

professionals include female athlete triad risk assessment in the standard PPE and consider expanding the history to include full menstrual history, reasons for hormonal therapy use, and questions concerning eating disorders and eating irregularities.⁵⁶

Dermatology

Athletes should be asked about a history of dermatologic pathology, with particular attention to highly communicable infections such as herpes or methicillin-resistant *Staphylococcus aureus*.² The athlete should be inspected for evidence of common cutaneous infections such as herpes gladiatorum, tinea gladiatorum, impetigo, molluscum contagiosum, warts, and community-acquired methicillin-resistant *S. aureus*.² Prevention of transmission is critical and may be achieved by covering the infected site, using prophylactic medications as indicated, refraining from sharing personal items, thoroughly cleaning athletic equipment, or ultimately restricting the athlete from participation for a specific period based on the characteristics of the sport, the type of microorganism involved, and governing body guidelines.²

Immunizations

Lack of immunizations does not inherently affect sports participation, but many states require them for school enrollment, and thus the PPE provides a good opportunity to discuss vaccinations.² Athletes traveling internationally for competition should be aware of local immunization guidelines recommended by the Centers for Disease Control and Prevention.²

Medications and Allergies

It is imperative that medical personnel are familiar with regulations established by drug-enforcing agencies such as the World Anti-Doping Agency. Some medications are strictly prohibited, whereas others, such as albuterol, may be used with therapeutic use exemption.⁵ Over-the-counter drugs and supplements may contain banned substances and should also be thoroughly reviewed. The PPE provides a good opportunity to address the detrimental effects of illicit drug and alcohol use, particularly in the adolescent population. Medication, food, and environmental allergies should be documented in detail. Specifications should include the name of the allergen, the type of reaction, and whether athletes require an epinephrine autoinjector.² Four recommendations for persons with a history of anaphylaxis include:

1. All medical personnel and athletes with allergies who may potentially require treatment with an epinephrine autoinjector should be trained in how to use it.
2. All medical kits should be stocked with an epinephrine autoinjector and over-the-counter diphenhydramine.
3. Athletes with allergies should carry an epinephrine autoinjector in their backpacks, have an additional one in their homes or dormitory rooms, and have over-the-counter diphenhydramine readily available at all times.
4. An emergency action plan (EAP) should be detailed for athletes with allergies during the PPE.

Heat Illness

Athletes with a history of exertional heat illness (EHI) are at enhanced risk for recurrent heat illness.⁵⁷ Such athletes should be

educated about preventive measures, including adequate hydration and gradual acclimatization over a period of 10 to 14 days. If possible, use of stimulant and antihistamine medications should be avoided during warm-weather activities.⁵ Athletes who use stimulant medications (e.g., for attention-deficit/hyperactivity disorder) and team health care professionals should be informed about the possible deleterious effects of using these medications in hot environments.

The National Athletic Trainers' Association (NATA) has guidelines for athletes to help mitigate risks involved with EHI.⁵⁸ It is recommended that individuals are gradually acclimatized to heat over 7 to 14 days to prevent the risk of EHI. Athletes who are currently sick with any viral infection or fever or have a skin rash should be withheld from participation until the condition is resolved. It is recommended that players have access to fluids at all times. Athletes should consume sodium-containing food or fluids to help replace loss of sodium in sweat and/or urine. Monitoring proper fluid consumption and replacement can help to minimize body mass loss to 2% during activity participation, which should be measured before and after the activity. Athletes should also be encouraged to sleep a minimum of 7 hours per night in a cool environment, eat a balanced diet, and properly hydrate before, during, and after exercise. Of most importance, rest periods should consist of at least 3 hours for food, nutrients, and electrolytes to be properly digested and absorbed before the next practice or game. Cold-water or ice towels should be readily accessible for all patients because immediate whole-body cooling is essential to treat EHI.

Athletes With Special Needs

The PPE-4 includes a chapter that discusses athletes with special needs and a separate form to aid the provider with unique issues affecting this patient population. The history and physical examination should be similar to that used with athletes who do not have special needs but may need to focus on diseases more common in this population such as seizures, hearing loss, vision loss, congenital heart disease, and renal disease.² Athletes with Down syndrome should always be assessed for a history of atlantoaxial instability (AAI), and athletes with spinal cord injuries should be asked about difficulties with thermoregulation, autonomic dysreflexia, pressure ulcers, and use of urinary catheterization.² The physical examination should focus on the visual, cardiovascular, musculoskeletal, neurologic, and dermatologic systems. Congenital heart disease is present in up to half of all athletes with Down syndrome and may require a cardiology referral for further testing prior to participation in sports.⁵⁹

The neurologic examination in athletes with Down syndrome should include evaluation for symptomatic AAI, which may present with upper motor neuron signs such as spasticity, hyperreflexia, clonus, and clumsiness. Persons with symptomatic AAI should undergo lateral cervical spine flexion and extension plain x-ray views to assess stability.² Wheelchair athletes should be evaluated for nerve entrapments, pressure ulcers, and overuse injuries of the shoulders, hands, and wrists.²

Conclusion

The PPE has continued to evolve, with updated recommendations to perform a PPE for all children in an office-based setting. Despite these broad recommendations, evidence showing that the PPE reduces morbidity and mortality is scarce.⁴ The PPE-4 emphasizes that history, rather than physical examination, is most sensitive for gleaning information relevant to medical clearance for sports participation. Nonetheless, a thorough PPE using a systematic approach and current guidelines may help to identify those athletes who need additional investigation prior to safe athletic participation and may also aid in preparation by the sports medicine team and the athlete for any injuries or medical complications that may result from sports.

ON-FIELD EMERGENCIES

The preceding section outlined the limitations in preventing potentially life-threatening events. Although rare, medical emergencies in sports will inevitably occur despite thorough preventive measures. This section outlines the core components of managing sports-related emergencies, including development of an effective EAP, the general approach to the collapsed athlete, and select examples of medical emergencies encountered on the playing field.

Emergency Action Plan

Before hosting athletic events, every institution has a duty to formulate an EAP,⁶⁰ which should be developed and reviewed annually as a coordinated effort by the medical team, school administrators, local emergency medical services (EMS), and first responders. The key components of the EAP are (1) effective communication, (2) training personnel, (3) acquiring equipment, (4) emergency transportation, (5) practice and review, and (6) postevent guidelines.⁶¹

Effective communication with local EMS is a critical link in the chain of survival. Communication systems such as telephones or two-way radios should be tested before each game. An EAP coordinator should be identified at each institution, and all other potential first responders should receive training in EAP guidelines, as well as in CPR and AED use. Resuscitation equipment should be centrally located and well marked for easy access. Personnel trained in basic life support (BLS) should have quick access to barrier masks for rescue breathing and an on-site AED. Providers trained in advanced cardiac life support (ACLS) may consider having advanced airways, oxygen delivery systems, and cardiac medications available.⁶¹ Other critical contents of a sideline physician's emergency kit are listed in [Box 11.4](#). A plan for efficiently directing EMS to the field of play should be in place, and on-site ambulance coverage is ideal at particularly high-risk events. In the event that medical personnel must leave the playing field to tend to an injured athlete, a plan for continued event coverage should be established.⁶⁰ The EAP should be rehearsed on an annual basis, which will ensure a coordinated team effort, and the EAP should be evaluated frequently to ensure its durability. Lastly, a protocol for release of medical information should be established and an incident form should be available for documentation and evaluation purposes.⁶¹

BOX 11.4 Recommended Contents of the Sideline Physician's Emergency Bag**Medications**

- Aspirin
- Beta-agonist inhaler
- Epinephrine
- Oral glucose solution

Wound Care

- Bandage scissors
- Bandage tape
- Sterile saline and sterile bowl (0.5–1 L)
- 60-mL syringe for irrigation
- Large-bore needles: 18 gauge or angiocatheter
- Antibacterial cleansing agent
- 1% lidocaine with and without epinephrine
- Lidocaine jelly (2%) or LET gel for abrasions
- Sterile scrub brush
- Povidone-iodine
- 5-mL syringes
- 25-gauge needles (1 Y2-in. long)
- Needle driver
- Tweezers

- Skin stapler
- Sterile drapes or towels
- Nylon sutures: 3-0, 4-0, 5-0, 6-0
- Polyglactin sutures: 3-0, 4-0, 5-0
- Liquid cyanoacrylate adhesive
- Compound benzoin tincture
- Bacitracin or triple antibiotic ointment
- Adhesive strips: Y4-, Y2-, and 1-in. wide
- Nonadherent pads
- Sterile gauze pads
- Sterile adhesive dressings in various sizes
- Hydrocolloid dressings
- Protective skin covering
- Aluminum chloride solution
- Athletic tape
- Latex and nonlatex (e.g., nitrile) gloves
- Aftercare instructions for patients

Airway Devices

- Pocket mouth-to-mouth mask
- Cutting tool for facemask loops

Modified from Daniels JM, Kary J, Lane JA. Optimizing the sideline medical bag preparing for school and community sports events. *Phys Sportsmed.* 2005;33(12):9–16; Bouchard M. Sideline care of abrasions and lacerations: preparation is key. *Phys Sportsmed.* 2005;33(2):21–29.

TABLE 11.2 Initial Approach to the Collapsed Athlete**Nontraumatic****BLS Assessment⁴⁹**

1. Check for responsiveness.
2. Activate the emergency response system; obtain and activate the AED.
3. Check for breathing and a carotid pulse for 5–10 s. If no pulse is present, initiate CPR with chest compressions first and then rescue breathing. If a pulse is present, begin rescue breathing.
4. Perform defibrillation with the AED when indicated.

ACLS Primary Assessment⁴⁹

Airway: Maintain airway patency and use advanced airway management if needed.
Breathing: Monitor ventilation and oxygenation and provide supplementary oxygen if needed.
Circulation: Monitor CPR quality; monitor for arrhythmias; provide defibrillation, cardioversion, and administer cardiac drugs when needed; obtain intravenous or intraosseous access and provide fluids if needed; monitor glucose, temperature, and perfusion.
Disability: Check neurologic function; check for responsiveness, levels of consciousness, and pupil dilation.
Exposure: Remove clothing and observe for signs of trauma, bleeding, or medical alert bracelets.

Traumatic**ATLS Primary Survey⁴¹**

Airway maintenance with cervical spine control
Breathing and ventilation
Circulation and hemorrhage control
Disability/drugs: assessment of neurologic status
Exposure: completely undress the patient but prevent hypothermia

ATLS Secondary Survey⁴¹

History:
Allergies
Medications
Past illness/Pregnancy
Last meal
Events/Environment related to injury
Physical examination:
 Thorough head-to-toe physical examination

ACLS, Advanced cardiac life support; AED, automated external defibrillator; ATLS, advanced trauma life support; BLS, basic life support; CPR, cardiopulmonary resuscitation.

General Approach to the Collapsed Athlete

A pivotal duty of physicians is to be a keen observer of on-field emergencies, because the injury mechanism is often critical to the initial management of the athlete. For example, the clinical approach to a 65-year-old masters track athlete who clutches his chest before collapsing in the middle of a race will differ from the care of a 15-year-old football cornerback who collapses after “spear tackling” an opponent. In the first nontraumatic case, cardiac etiology would be suspected, and thus the BLS and

ACLS assessments would be used (Table 11.2). Conversely, the approach to the trauma patient should follow the advanced trauma life support (ATLS) guidelines (see Table 11.2).⁶² With ATLS, the rapid primary survey assesses vital functions with simultaneous stabilization of any life-threatening conditions. After stabilization, a thorough history and physical examination of the secondary survey should commence.⁶² Physicians covering a sporting event may not always be able to determine the injury mechanism or extent of the injury, particularly when the patient remains unconscious or the inciting event was not witnessed.

In these situations, it should be assumed that the cervical spine is compromised, and physicians should follow appropriate precautions (outlined later in this chapter).

Head Injury

Head injuries account for 69% of fatalities in American football athletes.⁶³ The leading cause of death after a sports-related head injury is intracranial hemorrhage,⁶⁴ which may present as either an epidural, subdural, intracerebral, or subarachnoid hematoma. Epidural hematomas, which typically result from a tear of the middle meningeal artery as a complication of fracture of the temporal bone, are the most rapidly progressive intracranial hemorrhage and may cause death within 30 to 60 minutes.⁶⁴ In contrast, subdural hematomas evolve at a considerably slower rate, over a period of days to months, and yet they are the most common cause of fatal head injury in athletics.⁶³ Fortunately, most head injuries are less severe, with concussions being more common. A concussion is defined as a traumatic brain injury induced by direct or indirect biomechanical forces.⁴¹ The sideline physician should be trained to recognize the symptoms of concussion, which vary greatly and include combinations of different symptoms including but not limited to headaches, nausea, dizziness, photophobia, and difficulties with concentration and memory.⁶⁵

The approach to a collapsed athlete with a suspected head injury should always begin with the ATLS survey. An unconscious athlete should be assumed to have a cervical spine injury and should be appropriately stabilized. For conscious athletes, the primary survey should involve rapid assessment of circulation, airway, and breathing, followed by an assessment for spinal tenderness and neurologic assessment of the upper and lower limbs.⁶⁵ The athlete who remains unconscious for greater than 1 minute, has focal neurologic deficits, or has deteriorating mental status should be transferred emergently for advanced imaging to assess for intracranial hemorrhage. Stable athletes should be removed from play, and a thorough neurologic examination should be performed either on the sideline or in the locker room. The SCAT⁴¹ provides a systematic approach to concussion evaluation both immediately on the field and for more extensive evaluation in the locker room, training room, or medical office. Following a potential head injury during athletic participation, serial examinations are essential because epidural hematomas frequently present with a lucid interval prior to rapid neurologic deterioration.⁶⁴ The concussed athlete should never return to play (RTP) the same day and should follow a gradual stepwise return-to-sport protocol, returning to play only after symptoms have resolved and medical clearance has occurred.⁴¹ Current guidelines no longer allow for “rare exceptions” to be made with respect to same day RTP guidelines for elite adult athletes. We recommend that team physicians, in concert with athletic trainers and other medical personnel involved in the care of athletes, establish a written policy regarding concussion management. The policy should detail a stepwise algorithm to carefully assess clinical progress, cognitive function (using computerized testing when available), a graded return-to-sport protocol, and, as needed, a plan for specialist consultation.

Cervical Spine Injury

Approximately 12,000 new spinal cord injuries occur each year in the United States, with 8% resulting from sports participation.⁶⁶ The injury typically results from traumatic axial loading of the cervical spine, which creates transient, geometric changes in the spinal column with resultant injury to neural structures.⁶⁷

The recommendations provided in this section are primarily based on the NATA position statement on prehospital management of the spine-injured athlete. At the time of press, the NATA was completing a position statement update process, and the recommendations herein are reflective of changes outlined in the corresponding executive summary; however, these updates had not been officially endorsed by the NATA and participating professional organizations. The authors encourage the reader to review the NATA website for updates (<https://www.nata.org/news-publications/pressroom/statements/consensus>).

The management of suspected cervical spine injury requires a coordinated team effort, and prior to each season, neck stabilization techniques, methods of transferring injured players, airway and equipment management, and immobilization techniques should be rehearsed carefully. EAPs should be developed and coordinated with local emergency responders prior to the season, and on-site sports medicine teams should perform a “time out” prior to competition to review EAPs. Considerations should include the personnel and equipment available for that event, stabilization and transportation plans, and the use of a hospital that can deliver immediate, definitive care for cervical spine injuries.⁶⁸

In a collapsed athlete, symptoms and signs that raise the index of suspicion for severe traumatic cervical spine injury include (1) unconsciousness or an altered level of consciousness, (2) bilateral neurologic findings or complaints, (3) significant midline spine pain with or without palpation, and (4) obvious spinal column deformity.⁶⁹ An ATLS survey should commence immediately with special attention to cervical spine precautions. Moving the patient to a supine position with use of the log-roll maneuver may be obligatory if the athlete is found prone and the primary survey cannot be performed. The head and neck should be manually maintained in a neutral position without traction.⁶⁹ Airway patency and accessibility should be established, and rescue breathing should be initiated if necessary, which may entail the removal of a facemask or mouth guard. A cordless screwdriver is the initial equipment of choice for removal of a facemask because it induces less movement than other tools. However, an appropriate manual tool for cutting helmet loops should also be available at all times, and medical personnel should rehearse the use of these tools before the season begins. Helmet screws also should be routinely inspected and maintained throughout the season by equipment managers and/or athletic trainers. Removal of helmets and pads prior to transfer when appropriate should be done by three rescuers trained and experienced with equipment removal, or at the earliest possible time three such trained rescuers are available in order to have unhindered access to the athlete to facilitate delivery of care.^{68,69} Manual stabilization should be augmented, but not replaced, by mechanical stabilization with use of a cervical collar. A full-body stabilization should be

facilitated by transfer to a spine board. Supine athletes should be transferred via the “lift-and-slide” technique, whereas prone athletes should be transferred with use of the “log-roll” method. Head and neck stabilization should continue during transportation and until a destabilizing injury has been excluded. For an in-depth review, we recommend perusal of the NATA position statement⁶⁸ on acute management of the cervical spine-injured athlete.⁶⁹

Cardiac Arrest

The most common cause of SCD in athletes younger than 35 years is congenital cardiac disease, and in athletes older than 35 years, the most frequent etiology is atherosclerotic heart disease.⁷⁰ The initial presenting feature of occult cardiac disease may be SCD, and thus the PPE is designed to screen for potential symptoms and signs that may be representative of underlying cardiac disease. In the event of a nontraumatic collapse of an athlete, a potential cardiac etiology should be strongly considered and a thorough evaluation and cardiology consultation are imperative before the athlete is allowed to resume participation. After cardiac arrest, the inclusive goal is to support and restore effective oxygenation, ventilation, and circulation with return of neurologic function.⁷¹ Keys to ensuring successful resuscitation are (1) immediate recognition of cardiac arrest and activation of the emergency response system, (2) early initiation of CPR with an emphasis on high-quality chest compressions, (3) rapid defibrillation, (4) effective use of advanced life support measures, and (5) integrated post-cardiac arrest care.⁷¹

Initial evaluation should follow the BLS assessment and ACLS primary assessment (see Table 11.2). For unconscious athletes with cardiac or respiratory arrest, initial management should move to the ACLS primary assessment once the BLS assessment is completed. For conscious athletes, the ACLS primary assessment should be conducted first.⁷¹ The 2015 AHA BLS guidelines recommend beginning CPR with chest compressions immediately, without a delay for administering rescue breaths first. Factors for effective CPR include a rate of 100 to 120 chest compressions per minute, adequate depth of compressions to 5 cm, allowing complete chest recoil after each compression, minimizing interruptions of chest compressions, avoiding excessive ventilation, and switching the CPR provider who is administering the chest compressions to avoid fatigue, ideally changing positions during cardiac rhythm interpretation by the AED, as applicable.⁷¹ At a minimum, we recommend that all medical providers involved in the care of athletes maintain BLS certification, and ACLS certification for team physicians should be strongly considered.

Tension Pneumothorax

Pneumothorax (PTX) is the presence of air in the pleural space, which occurs spontaneously as a result of a traumatic tear of the pleura after a chest injury or may be iatrogenic as a result of a medical procedure.⁷² Tension PTX is a rare form of PTX that may develop as a result of a chest wall defect or a displaced fracture, or as a progression of a simple PTX. The underlying mechanism is a “one-way valve” air leak that forms through a defect in the

lung or chest wall, which leads to a progressively enlarging PTX with each inspiration.⁶² Typical symptoms include dyspnea and ipsilateral pleuritic chest pain. Physical examination may reveal tachycardia, tachypnea, hypotension, hyperresonance to percussion, unilateral absence of breath sounds, tracheal deviation, and distended neck veins.⁷² Sideline diagnosis of this medical emergency is dependent on symptoms and signs noted with clinical evaluation, and clinicians should not await radiographic imaging to confirm this diagnosis.⁶² Initial management entails immediate decompression via insertion of a large-caliber (11- to 16-gauge) needle into the affected hemithorax by way of the second intercostal space in the midclavicular line.⁶² Definitive treatment involves emergent hospital transportation, tube thoracostomy, and supplementary administration of oxygen. RTP guidelines should be individualized, yet it is possible for athletes to resume participation 3 to 4 weeks after a traumatic PTX has occurred.⁷² We recommend a thorough workup for occult connective tissue disorders (e.g., MFS) in athletes who experience a spontaneous PTX.

Anaphylaxis

Anaphylaxis is an acute, alarming, and accelerated life-threatening systemic reaction with varied mechanisms, clinical presentations, and severity that results from the abrupt systemic release of mediators from mast cells and basophils.⁷³ The overall incidence is estimated to be 1%–2% worldwide, and food and drugs are thought to be the most common causes.⁷⁴ Exercise-induced anaphylaxis is a well-described phenomenon in which exercise triggers symptoms, including extreme fatigue, warmth, flushing, pruritus, and urticaria, occasionally progressing to angioedema, wheezing, upper airway obstruction, and collapse.⁷³ The precise pathophysiology is ill defined, yet co-triggers such as nonsteroidal antiinflammatory drugs, pollen, or specific foods have been implicated and should be identified if possible.⁷⁵ Anaphylaxis as a result of insect stings is also a concern; symptoms are similar to those previously described but may also include a large, localized skin reaction.⁷³

The circulation, airway, breathing, and level of consciousness of athletes with a possible anaphylactic reaction should be evaluated. EMS should be activated followed by the rapid administration of an age-appropriate dose of epinephrine intramuscularly into the anterolateral thigh. Epinephrine injections can be repeated every 5 minutes for protracted symptoms. It is important to note that epinephrine injections alone do not definitively treat anaphylaxis in all cases and further emergent care may still be indicated. Athletes should be placed in a supine position with their lower limbs elevated. Vital signs should be monitored diligently, and oxygen and intravenous fluids should be administered if available. An advanced airway may be necessary for persons with respiratory compromise. Other adjunctive medications include nebulized bronchodilators, antihistamines, and glucocorticoids. After stabilization, athletes should undergo a workup to identify possible triggers of anaphylaxis, and future prevention should include appropriate prophylactic medications and avoidance of triggers. An epinephrine autoinjector should be prescribed, and a personalized EAP should be developed (as detailed in the PPE section).⁷³

Limb-Threatening Injuries

Knee Dislocation

Defined as complete disruption of the integrity of the tibiofemoral articulation, knee dislocation is typically accompanied by rupture of at least two of the four major knee ligaments.⁷⁶ Sports activities account for one-third of all cases of knee dislocation.⁷⁷ The most feared complication is potential limb amputation as a result of vascular compromise. Dislocations are classified according to tibial dislocation in relation to the femur. Anterior dislocation as a result of hyperextension is the most common type of dislocation.⁷⁸ Clinical signs may be subtle if the dislocation reduced spontaneously, yet hemarthrosis and severe multidirectional instability are suggestive of a possible dislocation. If the knee remains dislocated, closed reduction should be attempted emergently. Common fibular nerve and popliteal artery function and integrity should be tested and documented. Distal pulses may remain intact despite significant popliteal artery compromise, which necessitates more definitive evaluation with angiography.⁷⁹ The athlete should be emergently transported for orthopedic and vascular surgery consultation.

Traumatic Amputation

The primary clinical goals after a traumatic amputation occurs are hemorrhage control and digit/limb salvage. Even small digital amputations may result in significant bleeding. Pressure should be applied to the area, and the affected limb should be elevated. A tourniquet should be applied only if significant hemorrhaging continues. The residual limb should be rinsed with sterile saline solution and wrapped with sterile gauze. The amputated portion should be wrapped in moist sterile gauze, sealed in a plastic bag, placed in an ice bath, and sent with the athlete during emergency transport to the hospital.⁸⁰

Acute Compartment Syndrome

Acute compartment syndrome (ACS) is caused by altered tissue homeostasis within a closed muscle compartment. Tissue pressure greater than vascular perfusion pressure reduces capillary blood flow and induces local tissue hypoxia and necrosis.⁸¹ ACS can occur in any limb but is much more common in the lower limbs. Causes of ACS in sports include fractures and crush injuries. Symptoms typically present within a few hours of the initial injury and include progressive pain, paresis, and paresthesias.⁸¹ Passive and active ROM worsen pain, and severe cases present with diminished pulses, pallor, and paralysis.⁸² Clinical suspicion for ACS mandates rapid transportation to a hospital for an emergent fasciotomy.

Environmental Factors

Heat Illness

EHI includes heat cramps, heat exhaustion, and heat stroke. Exertional heat stroke (EHS), the most severe type of heat illness, is defined by a core temperature greater than 40.5°C (105°F) at collapse, accompanied by central nervous system changes.⁸³ EHS manifests when body organ temperature rises above critical levels, leading to cell membrane damage and cell energy disruption. Early symptoms of EHS include clumsiness, stumbling, headache, nausea, dizziness, apathy, confusion, and impairment of

consciousness.⁸³ Risk factors include obesity, prior heat stroke, dehydration, certain medications, poor fitness, hot and humid environments, and lack of acclimatization.⁸³ Early recognition and treatment are fundamental to reducing morbidity and mortality. Circulation, airway, and breathing should be assessed, along with rectal temperature in athletes exhibiting signs of EHS. The goal is to lower core body temperature to less than 38.9°C (102.5°F) within 30 minutes of collapse. Confirmation of EHS should result in initiation of immediate cooling measures, ideally in the form of whole-body ice-water immersion, which should be performed *prior* to emergency transportation.^{83,84} The water should be approximately 1.7°C (35°F) to 15°C (59°F) and stirred continuously to maximize cooling. The patient should be removed from cold water immersion once core body temperature reaches 38.9°C (102°F) to prevent overcooling. Establishing intravenous access with volume resuscitation may help to prevent cardiovascular collapse and end-organ damage. A rectal temperature should be obtained every 3 to 5 minutes, and cooling should continue until body temperature is less than 39°C.⁸⁵ All persons experiencing EHS should be transported to a medical facility for advanced care and monitoring.

Lightning

Injuries from lightning kill approximately 24,000 people annually worldwide.⁸⁵ Participants in outdoor recreational activities have the greatest risk of sustaining an injury from lightning.⁶² Lightning can cause injuries via direct contact, contact with an object that is struck, or “side splash,” which occurs when the electrical current jumps from an object to the victim.⁸⁵ The most common cause of fatality resulting from a lightning injury is cardiopulmonary arrest.⁸⁶ Other sequelae may include burns, fractures, and dislocations resulting from violent muscle contraction and neurologic conditions such as loss of consciousness, seizure, headache, paresthesia, confusion, memory loss, and transient paralysis.^{85,86}

Initial management may require movement of the injured person if ongoing thunderstorms threaten the safety of first responders. It should be noted that after a lightning strike, the injured person does not retain an electrical charge, and thus treatment should not be delayed.⁸⁷ After the scene of the lightning strike is deemed secure, management should follow the ATLS primary and secondary surveys. A tenet of triage after lightning trauma dictates that the “apparently dead” should be resuscitated first. This tenet pertains to the unique pathophysiology of cardiopulmonary arrest in lightning strikes that necessitates immediate ventilatory support to prevent secondary hypoxic arrhythmias.⁸⁶

Conclusion

Paramount to the management of on-field emergencies are the precepts of preparation and anticipation. Team physicians should coordinate EAPs and have them rehearsed annually by all staff involved in the care of athletes. In addition, the physician's kit should be restocked adequately on a periodic basis to help ensure effective and timely treatment. Physicians should remain alert on the sideline, monitoring play vigilantly so they can view injuries as they occur, which will facilitate optimal care.

Lastly, thorough assessment of injured and ill athletes can be enhanced by meticulous adherence to the BLS, ACLS, and ATLS guidelines.

ETHICAL AND LEGAL ISSUES IN SPORTS MEDICINE

Ancient Greeks were credited with the birth of both rational medicine and professional sports.⁶⁵ With Western medicine in its infancy, the physician's role in the sporting microcosm was neither well defined nor respected. Greek athletes sought advice from their trainers regarding their inclusive training methods, including diet and physiotherapy.⁸⁸ The awkward inception of medicine in the sporting arena was broadened by the sometimes contrasting goals of the athletes and physicians. Although the health of the athletes takes precedence for physicians, athletes often view victory as the ultimate goal.⁸⁸ Some persons believe that it is these opposing philosophies that lie beneath most ethical and legal issues in modern-day sports medicine. In addition, the evolution of the traditional dyad of the patient-physician relationship into a triad of patient-team-physician has been further recognized as a source of potential medicolegal conflict.⁸⁹ This section is intended to serve as a general guide for ethical and legal issues and should not be perceived as binding legal advice. For detailed advice regarding specific medicolegal issues, we recommend that legal counsel be obtained.

Confidentiality

Inherent to the integrity of any patient-physician relationship is the concept of trust. In modern medicine, a primary approach to maintain the trust of our patients is to preserve confidentiality. The following case example illustrates potential problems surrounding trust.

Case 1

A 21-year-old star running back cuts to the sideline during a midseason game. He hears a "pop" in his right knee, which is accompanied by immediate pain and swelling. As the team physician, you evaluate him, and examination reveals positive right knee Lachman and anterior drawer tests. Immediately following your examination, you are approached by the athletic trainer, the head coach, and a member of the media who ask for return-to-play prognosis. During this encounter, you have not asked the player for his consent to release medical information.

Question: Who can you talk to, and what can you tell them?

Many clinicians reflexively cite confidentiality guidelines outlined in the Health Information Portability and Accountability Act (HIPAA). Remarkably, the primary goal of HIPAA was not to enforce confidentiality but rather to improve the portability and continuity of health insurance coverage.⁹⁰ A section of HIPAA details the principles of patient confidentiality and subsequently had far-reaching effects on patient information when it was enacted in 2003. A key determinant to the enforcement of HIPAA guidelines depends on whether the information is "protected." These regulations apply only to clinical encounters for which physicians bill for their services, receive

payments, or transmit claims to a health plan in an electronic form.⁹⁰ Regardless of the clinical scenario, revealing whether an athlete is "cleared" or "not cleared" is considered an exception to HIPAA.⁹⁰

High School Athletics

At the high school level, with rare exception, sideline coverage is voluntary and does not involve any form of billing, and thus the information obtained during these encounters would not be considered under the jurisdiction of HIPAA. Theoretically, information could be shared with coaches and team administrators without fear of prosecution from the Department of Health and Human Services, which implements HIPAA. Although no legal precedent exists, some persons believe that HIPAA guidelines should be adhered to in high schools.⁹⁰ Regardless, if medical services are billed, for example, as an in-office follow-up visit, HIPAA guidelines are enforceable.

Collegiate Athletics

HIPAA guidelines are usually the rule in collegiate athletics. One particular exception is clinical scenarios that involve health information governed by the Federal Education Records Protection Act (FERPA). At private or public colleges and universities that receive federal educational funding, information obtained during clinical encounters may be covered by this act. Information governed by FERPA guidelines are exempt from HIPAA and enable health information to be released to administrators and coaches who have an educational interest in the athlete without formal consent by the patient.⁹⁰ School-based athletic training room records maintained by team physicians or certified athletic trainers are often considered to be governed by FERPA. Given the complex nature of FERPA and HIPAA regulations, it is advisable to review privacy policies with the institution's legal counsel.²

Professional Athletics

In the arena of professional athletics, athletes sign an employment contract with the team, and health information can be considered to be a part of the employment record. Thus health information gathered by a team physician may be shared with employers who have a vested interest in the athlete's health, such as coaches and administrators.⁹⁰ As previously illustrated, the issue of whether information can be divulged to coaches varies with the clinical situation. To avoid controversy when dealing with sensitive medical information, most professional teams have their athletes sign a preparticipation waiver that details with whom health information can be shared as needed. Thus consent for release of information does not have to be obtained repeatedly throughout the season. In the absence of such a waiver, if an athlete begins to reveal information to the team physician that could potentially restrict him or her from play, it is recommended that the physician preempt any further revelations by notifying the athlete that if further information is shared, he or she may need to share this information with the team's administration. At that juncture, the athlete can consent to the release of medical information or refuse to reveal additional information and request a referral to another provider.⁹¹

TABLE 11.3 Key Components of Informed Consent

Component	Description
Disclosure	Relevant information must be provided regarding the proposed treatment including potential risks, benefits, and alternatives.
Capacity	Athletes must have the ability to understand information and its relevance.
Voluntariness	Athletes must voluntarily express their wishes in a noncoercive environment.

Informed Consent

Informed consent is one of the primary means of ensuring patient autonomy when making nonemergent medical decisions.⁹² Most physicians are familiar with the concept of informed consent in the context of preprocedure planning, but any medical decision affecting an athlete's health and future should involve three key components of informed consent: disclosure, capacity, and voluntariness (Table 11.3).⁹³ To illustrate the importance of informed consent, we provide the following case.

Case 2

A 20-year-old female middle distance runner reports recurrent right medial ankle pain and swelling consistent with a previous diagnosis of posterior tibial tendinopathy. She is scheduled to participate in a regional qualifying meet for the national championships in 1 week. She requests a steroid injection into the posterior tibial tendon sheath and states that a prior injection by her primary care physician allowed her to successfully obtain a national qualifying result last year.

Question: Should you proceed with the injection or opt for more conservative measures?

In this example, pressure from the coach and athlete to do whatever it takes to win may affect clinical judgment by the team physician. Further complicating the scenario is the fact that she had a prior injection in the past with good results, and as such, it is reasonable for the coach and athlete to expect similar treatment. Referring to the basic components of informed consent (see Table 11.3), it is imperative to reestablish the three criteria in the current scenario because it is unlikely that the athlete can still recite the potential risks, benefits, and alternatives. It is also incorrect to assume that informed consent was actually obtained prior to her last injection.

The short-term, antiinflammatory, and analgesic effects of the injection would need to be weighed with the short-term risks of acute rupture and long-term risk of chronic weakening and progression of posterior tibialis tendon dysfunction.⁹⁴ By educating the patient about the risks, benefits, and alternatives, the physician is fulfilling a protective role in medical decision-making. This scenario does not involve a life-threatening injury; accordingly, it is reasonable that if informed consent is obtained, an injection may be justified. Conversely, in rendering return-to-play decisions for potentially catastrophic injuries such as cervical spine instability, patient autonomy would be less important than

the principles of beneficence and nonmaleficence. In addition, when athletes join a sports team, they voluntarily sacrifice a certain degree of their autonomy by agreeing to abide with the decisions of team physicians.⁹³

Another potential pitfall is the assumption that a signed consent form automatically fulfills the medicolegal criteria needed to substantiate informed consent. A legal precedent exists indicating that a signed consent form does not necessarily guarantee legal immunity. Medical jargon may be overly complicated, and the patient may not fully understand the potential risks of the procedure. It is recommended that if doubt exists about whether the patient fully understands the ramifications of a medical decision, the patient should write a letter in his or her own words detailing his or her understanding of the risks, benefits, and alternatives.⁹⁵

Drug Use

A trend in modern sports is the use of performance-enhancing drugs. In opposing the use of performance-enhancing drugs, most physicians cite the potential harms and the unfairness for the spirit of competition and good sportsmanship. The practice of physicians supplying banned substances to athletes is entirely unacceptable, and yet numerous well-publicized examples of such practices exist.⁹⁶ Although condemnation of the distribution of performance-enhancing drugs by a physician is clear-cut, other potential scenarios may be unclear.

Case 3

You are the primary care physician for an 18-year-old male sprinter who is competing on the junior national track team. He presents to your clinic with a report of acne. Upon further questioning, he reveals that he has recently started taking anabolic steroids.

Questions: Should this information be kept confidential? Should you continue to treat the patient? Would your decision change if you were the national team physician?

The physician may believe that it is necessary to inform his patient's coaches and potentially an antidoping agency that governs the sport. However, in this case, the athlete has addressed his use of steroids during a clinic visit with his primary care physician, who does not have a legal obligation to disclose this information. Some persons have suggested that athletes who use performance-enhancing drugs should not be viewed any differently from those who abuse alcohol or tobacco,^{96,97} because with the latter forms of abuse, physicians would provide counseling regarding potential adverse effects. Likewise, it is beneficial for physicians to advise athletes on the potential detrimental effects of performance-enhancing drugs, without overstating risks, to maintain their trust. In addition, psychosocial triggers that may have fueled drug use should be considered and addressed. On the other hand, in this scenario, if the physician was taking care of an athlete on the national team rather than serving as a primary care provider, an obligation to divulge information to administrators is logical. However, in most instances at the national level, the athlete will have signed an agreement acknowledging forfeiture from participation for use of performance-enhancing drugs.

Disagreements With Coaches

Sideline medical coverage can exemplify the potentially incompatible goals of physicians and coaches. Occasionally, coaches may disagree with medical assessments, as illustrated in the following case.

Case 4

A 16-year-old football cornerback tackles an opposing player in the second quarter of play. You witness him jog back to the huddle with his right arm held at his side. He reports having radiating pain down his arm. A preliminary examination reveals no neck tenderness and $\frac{4}{5}$ shoulder abduction and elbow flexion strength. You instruct him not to RTP and tell him you plan to reexamine him in 10 minutes. The coach cannot understand why you have restricted play because the coach “used to play with stingers.”

Questions: What should you tell the coach? Could this situation have been avoided?

The RTP guidelines with regard to transient brachial plexopathies (also known as stingers) were different in the past, and athletes often returned to play relatively soon. However, current guidelines clearly state that RTP is contraindicated in the presence of residual neurologic deficits.⁴⁵ Knowledge of current guidelines and education of athletes and coaches can be of great help to team physicians.⁹⁸ A thorough explanation of the potential immediate and long-term sequelae of premature RTP should suffice for most coaches. Perhaps a solution to avoid a power struggle with a coach on the sideline is to preemptively delineate duties as team physician in writing. This approach typically would not be feasible in a situation in which a team physician is at odds with an opposing team's coach, especially when only one physician is on site at a game. In this situation, the physician should arrive early, introduce himself or herself to the opposing team's athletic trainer(s) and coaching staff, and delineate his or her role during the contest. To make unbiased medical decisions on either sideline, it is recommended that team physicians avoid excessive emotional involvement with their team and maintain a sense of objectivity throughout the contest.⁹³

Conclusion

The team physician encounters unique bioethical issues when interacting with coaches and athletes. Most of these issues stem from the nontraditional patient-doctor relationship and the dichotomous goals of “health first” versus “victory at all costs.” Team physicians should rely on their knowledge of current medical guidelines and the basic principles of medical ethics.

In addition, it is recommended that physicians strictly adhere to the rule of “full disclosure” and notify all athletes of potential risks and conflicts of interest, which will facilitate autonomous and informed decision-making.⁹²

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For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Bernhardt DT, Roberts WO, eds. *Preparticipation Physical Evaluation*. 4th ed. Elk Grove Village, IL: American Academy of Pediatrics; 2010.

Level of Evidence:

V, expert opinion

Summary:

A product of six author societies, the fourth edition of the preparticipation physical evaluation guides the team physician through the intricacies of preseason screening for sport participation.

Citation:

McCorry P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport – the 5th international conference on concussion in sport held in Berlin, October 2016. *Br J Sports Med*. 2017;[Epub ahead of print].

Level of Evidence:

V, expert opinion

Summary:

The most recent and widely accepted concussion management guideline, including in-depth discussion on RTP decisions.

Citation:

Swartz EE, Boden BP, Courson RW, et al. National athletic trainers' association position statement: acute management of the cervical spine-injured athlete. *J Athl Train*. 2009;44(3):306–331.

Level of Evidence:

V, expert opinion

Summary:

A consensus statement of the on-field management of the athlete with a cervical spine injury.

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